Morning Paper Report - Friday, October 16, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Optimized Temporal Scaffolding: Rhythmic Micro-Variations Selectively Enhance Visual Working Memory Precision

Xing Chen, Melinda Xiaoxiao Ying, Kaiwen Xiao, Qiurui Wang
EAI Endorsed Transactions on Pervasive Health and Technology, 2026

Source: Semantic Scholar; matched terms: visual working memory, working memory, fidelity, precision

Link: https://www.semanticscholar.org/paper/b508877793948c963284e4025f9fb8e87d6b04fd

Goal: Visual working memory (VWM) precision—the fidelity of stored visual representations—is a critical determinant of cognitive performance, yet the influence of subtle, non-semantic auditory features remains largely unexplored

Method: Participants performed the task under three auditory conditions: Silence, a Mechanically Isochronous Rhythm (ME), and a Micro-Variation Rhythm (MV) consisting of quasi-isochronous drum patterns at 240 BPM with ±20 ms timing variation

Results: These findings are consistent with the idea that the natural, non-linear temporal structure embedded in expressive rhythm can serve as an Optimized Temporal Scaffold for visual cognition, providing a more effective acoustic context than perfect isochrony for supporting high-fidelity VWM representations The work bridges research on musical expressivity and fundamental cognitive resource allocation and points to...

Conclusion: An exploratory analysis further suggested that the magnitude of the precision benefit was positively associated with participants' level of musical training

2. Examining the role of attentional allocation in working memory precision with pupillometry in children and adults.

Elena M. Galeano-Keiner, S. Pakzad, Garvin Brod, S. Bunge
Journal of Experimental Child Psychology, 2023

Source: Semantic Scholar; matched terms: working memory, fidelity, precision, modeling, mnemonic precision

Link: https://www.semanticscholar.org/paper/afb4253588b184ef825de9c8acf5d79e81f2cc0c

Goal: Working memory (WM) precision, or the fidelity with which items can be remembered, is an important aspect of WM capacity that increases over childhood

Method: Here, we examined the role of attentional allocation in visual WM precision in children aged 8 to 13 years and young adults aged 18 to 27 years, as measured by fluctuations in pupil dilation during stimulus encoding and maintenance

Results: We found an age-related increase in mnemonic precision that was independent of guessing behavior, serial position effects, fatigue or loss of motivation across the experiment, and visuomotor processes

Conclusion: These results suggest a functional link between pupil fluctuations and WM precision that grows over development; visual details may be stored more faithfully when attention is allocated efficiently to a sequence of objects at encoding and throughout a delay period.

3. Attractor dynamics underlie sequence working memory in data-constrained recurrent neural networks

Matthijs Pals, Jakob H. Macke
2026

Source: Crossref; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: https://doi.org/10.64898/2026.09.30.755565

Goal: Abstract Sequences of stimuli can be maintained in working memory by encoding each item, together with its position, as a distinct pattern of neural population activity

Method: Competing hypotheses about the underlying dynamics have been proposed: patterns of activity might be supported either by stable attractor states that persist over time, or by transient sequential dynamics To investigate the dynamics underlying sequence memory, we developed a framework for fitting low-rank recurrent neural networks (RNNs) to multi-session recordings of spiking activity, and applied it to 2,488...

Results: The RNNs enabled direct investigation of the underlying dynamics: By isolating the temporal and stimulus subspaces, we found a manifold of fixed points emerged during the memory retention period

Conclusion: Competing hypotheses about the underlying dynamics have been proposed: patterns of activity might be supported either by stable attractor states that persist over time, or by transient sequential dynamics

4. Study of cutting force predictability, signal complexity of different end milling CWE stages with different modelling methods

Hui Xie, Guang-Xian Li, A. Longstaff, S. Fletcher, Song-Lin Ding, W. Pan
The International Journal of Advanced Manufacturing Technology, 2024

Source: Semantic Scholar; matched terms: fidelity, precision, modelling, neural network, deep learning

Link: https://www.semanticscholar.org/paper/3ac3e1de3ecd2b7fa036217fab9fb8941cd33d7b

Goal: Cutting force analysis in milling processes is essential for precision metal cutting as it contributes to understanding tool wear, optimising machining performance, and ensuring overall process stability

Method: Cutting force analysis in milling processes is essential for precision metal cutting as it contributes to understanding tool wear, optimising machining performance, and ensuring overall process stability

Results: In contrast, the LSTM model showed the best prediction performance across all tested working conditions

Conclusion: Verification and analysis indicate that the cutting force predictability with FEM at “partial engagement” was the lowest, due to its lack of ability to describe TPF harmonics

5. Sensory Neural Noise as a Limiting Factor in Visual Working Memory Precision in Neurotypicals and Individuals with Schizophrenia

Woon Ju Park, Megan C. Ichinose, Gengshi Hu, Geoffrey F. Woodman, D. Tadin, Sohee Park
bioRxiv, 2025

Source: Semantic Scholar; matched terms: visual working memory, working memory, precision

Link: https://www.semanticscholar.org/paper/5756462635802361b8ba54b01c7f2ccec5612802

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.